

Speech-Pitch Detection Using Maximum Likelihood Algorithm

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Abstract: In this paper we present a robust algorithm for pitch detection and estimation. The algorithm is based on implementation of the maximum-likelihood estimation. The speech signal is digitized with 8 KHz, 10-bit Analog-to-Digital Converter (ADC) and the digitized data are stored in a PC where it is segmented into frames each of 32 msec. A theoretical analysis has been conducted in this paper to find the maximum-likelihood epoch determination signal that represent the Glottal Closure Instant (GCI) from which the period can be determined. Hilbert transform is incorporated into the algorithm to improve the overall performance of the algorithm. Utterances from three males and three females speakers have been used to test the algorithm. The algorithm has performed reliably and detected correctly voiced, unvoiced, and mixed modes of speech.

Introduction: Pitch is an important piece of information carried by the speech signal. Pitch estimation is essential for several speech analysis-synthesis applications such as speech computer-aided education for the deaf, speaker identification, speech recognition, and diagnosis of speech disorders. Pitch estimation is determined by the rate of the glottal pulse train. A robust pitch estimator can be based on a the maximum likelihood estimation of the period in the frequency domain. The goal of this study is to develop such robust algorithm that can estimate the pitch of different speakers and detect voiced, unvoiced, and mixed speech segments

Preprocessing of the Speech Signal: After the speech signal was recorded in an acoustically treated room, it was digitized, at 8 kHz, using a 10-bit analog-to-digital (A/D). The digitized data were normalized prior to computations. A Hamming window was used to segment the data into frames of 32 msec length and 7 msec frame overlapping. For each frame a vector of twelve LPC coefficients a_j was computed. The wavelet caused by an epoch can be expressed as

$$\hat{s}(n) = \begin{cases} \sum_{i=1}^p a_i \hat{s}(n-i) & 0 < n \leq \infty \\ G & n = 0 \\ 0 & \text{otherwise} \end{cases} \quad \dots(1)$$

where a_j are the LPC coefficients, G is a positive arbitrary constant and p is the order of the polynomial.

Maximum Likelihood Estimation: Given that $p(x/\alpha)$ is the conditional probability density function of a sample vector x with a parameter space α , then $p(x/\alpha)$ is called the likelihood

function. Maximum-likelihood estimation is the method in which if the vector x is known, the unknown parameter space α is estimated so that the likelihood function $p(x/\alpha)$ is maximized. Let $s(n + n_0)$ be the speech signal, where $n \in [0, N - 1]$ for N observations and n_0 is a sequence alignment delay. The difference between the speech signal, $s(n + n_0)$, and the wavelet caused by an epoch, $\hat{s}(n)$, is a Gaussian process, $x(n)$, with N independent dimensions and uniform variance, σ , that is

$$x(n) = s(n + n_0) - \hat{s}(n)$$

The conditional probability density function, or likelihood function, of this Gaussian process can be represented as

$$p(x/\alpha) = \frac{1}{(2\pi\sigma^2)^{N/2}} \exp \left\{ - \sum_{n=0}^{N-1} \frac{[s(n+n_0) - \hat{s}(n)]^2}{2\sigma^2} \right\}$$

where α is the parameter space corresponding to $[\sigma, a_1, \dots, a_p, n_0]$. The epoch occurs when the parameter space values are estimated to maximize the likelihood function.

The Algorithm: The algorithm is implemented in 8 steps: a) normalize the digitized speech data, b) compute the Linear Predictive Coding coefficients, c) apply auto-regression filter to generate an impulse-response sequence representing Equation (1), d) apply maximum-likelihood estimation e) apply Hilbert Transform to eliminate the weaker pulse, f) apply average-value subtraction to find a single peak, g) apply time- shift adaptation to the GCI signal before displaying the results, h) normalize the results and display them. Details of the algorithm can be found in [1].

Results: the algorithm was written in C/C++. Utterances from three male and three female speakers were used to test the algorithm. Each speaker was asked to speak five words - "one", "four", "six", "nine", and "zero" - with every word spoken ten times. In addition to the five words, five vowels were uttered by two speakers so that the algorithm could be tested for wholly voiced and therefore periodic signals. The algorithm detected the voiced, unvoiced and mixed frames and displayed the results correctly on the computer's screen.

Reference: [1] R. Adamjee, "Detection of the Glottal Closure Instant using the Maximum-Likelihood estimation Algorithm", Master Thesis, Southern Illinois University at Carbondale, 1994.